

Expected outcomes (max. 3,800 characters)

The project assembles a group of researchers with a proven record of collaboration, and it fosters new collaborations between several research professors. This includes new collaborative opportunities between Ravensbergen and Higgins, Mohamed and Pereira, Ravensbergen and Pereira, and all transportation scholars collaborating with De Barros, an expert in human rights and justice. The team takes previous collaborations and expands them in novel ways that will benefit the quality of the scholarship produced. Moreover, the codesign framework proposed builds on existing collaborations (e.g., between Co-PI Ravensbergen and Mohammed and Hamilton's transit agency) and fosters new connections with community stakeholders. The project aspires to empower users of transit to have a more fluid participation in decision-making processes that affect them, and to help providers of a key public service to better incorporate user knowledge. In this way, the codesign framework opens avenues for knowledge exchange between providers and users of transit and thus contributes to enhance the process and practice of effective governance.

Through this project, four graduate students and two post-doctoral researchers will be trained and gain various new skills, to the betterment of Canada's transportation and research expertise. Other outcomes of the project are of three kinds: communication, research materials, and tools. The project will produce ten peer-reviewed papers (1 article by each Masters student, 3 articles per Doctoral student, and 1 article by each post-doctoral researcher). In addition, there will be communications for lay audiences, in the form of case studies, policy briefs, and user's manuals. New data produced during the research will be shared in the form of open data products, which will be made available via web repositories in addition to registering them for digital object identification (doi) using Zenodo. Other research materials, such as research documents and computational notebooks, will be shared using public web repositories. The project aims to push the boundaries of current practices in open science by expanding on the current expertise of members of the research team. The project will also produce open-source tools (e.g., dashboards) to help stakeholders assess the equity implications of various transit provision scenarios. More generally, knowledge transfer will prepare the next generation of transportation equity champions.

With better knowledge, better choices will be facilitated for planners and policymakers at transit agencies and users of public transportation. Rather than basing decisions on relatively narrow fiscal considerations, transparent and publicly available tools will contribute to advance more equitable public transportation planning. As a matter of procedural justice in transportation, planning must consider all segments of society, and for participation to be meaningful relevant information must be accessible. The tools proposed contribute to fill an information gap about levels of service by clarifying in a more holistic way the distribution of transit's costs and benefits. The development of visual tools based on research that accounts for relevant perspectives of stakeholders will enhance public discourse, public policy, and planning. Generating knowledge about the equity implications of transit operations will help steer conversations from a predominantly economic perspective to a fuller accounting of transit's numerous non-monetary benefits. Generating the tools to implement this knowledge will help inform policy and practice related to urban planning, transport, social inclusion, and community development by translating abstract research into trustworthy, locally relevant, and actionable information.